FAKE NEWS DETECTION

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FAKE NEWS DETECTION USING MACHINE LEARNING

✲ ABSTRACT

In today's digital era, the spread of fake news has become a serious issue affecting society, politics, and decision-making. This project presents a Fake News Detection System that uses Machine Learning and Natural Language Processing (NLP) techniques to classify news headlines as real or fake. The system uses a labeled dataset, applies preprocessing techniques, and trains a classification model to make predictions. A web interface is built using Streamlit to allow users to input news headlines and get instant results. The project highlights the importance of data preprocessing, model selection, and handling dataset imbalance.

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1. INTRODUCTION

With the rapid growth of social media and online news platforms, misinformation spreads quickly. Fake news can mislead people and create confusion. Detecting fake news manually is difficult and time-consuming.

This project aims to automate fake news detection using Machine Learning. By analyzing textual patterns in news headlines, the system can classify whether a news item is real or fake.

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2. OBJECTIVES

- To build a machine learning model for fake news detection
- To apply Natural Language Processing techniques
- To develop a user-friendly web interface
- To improve model accuracy by handling dataset imbalance

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3. LITERATURE REVIEW

Fake news detection has been widely studied using various techniques:

- Machine Learning models like Logistic Regression and Naive Bayes
- Deep Learning models such as LSTM and BERT
- NLP techniques like TF-IDF and word embeddings

Naive Bayes is commonly used for text classification due to its simplicity and effectiveness.

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4. METHODOLOGY

4.1 Data Collection

The dataset was obtained from Kaggle and contains:

- News title
- Source information
- Label (Real/Fake)

4.2 Data Preprocessing

Steps performed:

- Removal of null values
- Conversion of text to lowercase
- Removal of special characters
- Text cleaning using regular expressions

4.3 Feature Extraction

Text data is converted into numerical form using:

- TF-IDF (Term Frequency – Inverse Document Frequency)

This helps in identifying important words in the dataset.

4.4 Handling Imbalanced Data

The dataset had more real news than fake news.
To solve this:

- Equal samples of real and fake news were selected
- Dataset was balanced to avoid bias

4.5 Model Selection

The following models were considered:

- Logistic Regression
- Naive Bayes

Final model used:

- **Multinomial Naive Bayes**

Reason:

- Works well for text classification
- Handles sparse data efficiently

4.6 Model Training

- Dataset split into training (80%) and testing (20%)
- Model trained using training data
- Predictions made on test data

4.7 Model Evaluation

Performance measured using:

- Accuracy
- Precision
- Recall
- F1 Score

4.8 Deployment

A web interface was developed using Streamlit:

- User enters news headline
- Model predicts whether it is fake or real
- Result displayed instantly

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5. SYSTEM ARCHITECTURE

User Input → Preprocessing → TF-IDF → Model → Prediction → Output

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6. IMPLEMENTATION

Tools Used:

- Python
- VS Code
- Streamlit

Libraries:

- Pandas
- NumPy
- Scikit-learn
- Pickle

Modules:

1. **model.py** – Training and saving model
2. **predict.py** – Loading model and predicting
3. **app.py** – User interface

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7. RESULTS

- Model successfully classifies fake and real news
- Balanced dataset improved prediction accuracy
- System provides fast and reliable output

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8. CHALLENGES FACED

- Dataset imbalance (more real news than fake)
- Model bias towards majority class
- Incorrect predictions initially

Solution:

- Balanced dataset
- Changed model to Naive Bayes
- Improved preprocessing

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9. FUTURE SCOPE

- Use deep learning models (LSTM, BERT)
- Analyze full news articles instead of headlines
- Add confidence score
- Deploy system on cloud

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10. CONCLUSION

This project demonstrates how Machine Learning can be used to detect fake news effectively. By applying NLP techniques and proper data handling, the system achieves good accuracy. The project also highlights the importance of dataset quality and model selection in building reliable AI systems.

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11. REFERENCES

1. Kaggle Fake News Dataset
2. Scikit-learn Documentation
3. Research papers on Fake News Detection
4. Online tutorials and documentation